INCH-POUND

A-A-52036
May 1, 1992

COMMERCIAL ITEM DESCRIPTION

LEVELS, SURVEYING, SELF-LEVELING AND TILTING

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-L-52918.

This description covers levels used for making differential elevation measurements in surveying and in other engineering applications.

Salient characteristics:

Design. The levels shall be as specified herein. The complete instrument shall consist of an internally focusing telescope capable of rotation in 360 degrees, an image lens system, crosshairs superimposed on the image (a reticle), a leveling system for the telescope, a telescope base capable of being physically fastened to a tripod, and a tripod. A sun shade, lenscaps and carrying case completes the basic system. The image formed by the lens system may be erect or inverted in the eye piece.

Classification: The levels shall be of the following types, styles, and categories as specified.

Type I - Tilting level.
  Style A - With micrometer eyepiece and plane-parallel attachment.
  Style B - Without optional micrometers and plane-parallel attachment.

Type II - Self-leveling level.
  Style A - Without horizontal circle.
  Style B - With horizontal circle for stadia and simple surveys.

Category 1 - 1st order accuracy: 1.54 seconds or less error.
Category 2 - 2nd order accuracy: 3.07 seconds or less error.

Beneficial comments recommendations, additions, clarifications, deletions, etc., and any data which may be of use in improving this document should be sent to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

AMSC N/A FSC 6675

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Type I level: A tilting level is one which must be manually leveled initially, and periodically manually checked for maintaining a level state.

Type I levels shall be provided with a circular spirit level on thebase of the instrument to be used for preliminary leveling. Type I levels shall also be equipped with a graduated differential leveling vial and a screw leveling system such that when the bubble is centered in the vial, the telescope is level. The instrument also has a prism system so that, after being approximately leveled using the circular spirit level on the instrument base, both ends of the bubble in the vial are observable through a separate eyepiece or through the object lens of the telescope as a split image. Precise leveling of the telescope is then achieved by adjusting the elevation screws so that the split bubble images appear to coincide.

Type I levels may, or may not, have an optical micrometer attached to a plate parallel to the plane of the instrument. The parallel plate optical micrometer permits the operator to measure horizontal distances with the instrument. An instrument which is equipped with a parallel plate optical micrometer is defined as type I, style A; if no micrometer is provided, the instrument is type I, style B.

Type II level: A self-leveling instrument is one which will automatically maintain the telescope in a level position once the instrument has been manually brought to an acceptable approximation of level. Type II levels are equipped with a pendulum type compensator for self-leveling capability and will also have a parallel plate optical micrometer. They may, or may not, be equipped with a horizontal circle mounted on the base for the purpose of measuring angles from zero through 360 degrees. An instrument without a horizontal circle is defined as type II, style A; if a horizontal circle is provided, the instrument is type II, style B.

Optical system. The optical system shall be aberration free, providing a clear, bright, high contrast, sharply defined image throughout the entire focusing range and field of view. Optical properties shall conform to the following:

1. Magnification, minimum - 30 diameters, ±5 percent,
2. Effective aperture of object lens, minimum - 39 mm,
3. Focus minimum - 12 feet,
4. Field of view, minimum - 60 minutes of arc,
5. Resolution - 4.0 seconds of arc or better,
6. Eye relief distance - not less than 7 mm.

Circular spirit level. The circular spirit level vial for a type I level shall have a sensitivity of 10 minutes of arc ±2 minutes per 2 mm division. The circular spirit level vial for a type II level shall have a sensitivity of 7 to 15 minutes of arc per 2 mm division. The circular spirit level viewer shall be attached to the telescope in a location permitting a view of the bubble that is eye-height, horizontal, and parallel to the line of sight.

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Dissimilar metals shall not be used in intimate 4.4.2 Dissimilar metals

Telescope level. The telescope level vial for type I levels shall have a sensitivity of 8 to 10 seconds of arc per 2 mm movement. The ends of the level bubble shall be visible at all temperatures from -25 to +120 °F.

Level vial sensitivity. The sensitivity of the level shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. The plane of the graduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble, as indicated on a scale placed adjacent to the vials.

Support. The level shall be mounted on a tribrach. The tribrach shall consist essentially of a baseplate and three leveling screws. The tribrach baseplate shall have at its center a 5/8-inch nut of unified coarse thread series, (5/8-11-UNC-2B), class 2B, 11 threads per inch, as defined in FED-STD-H28. The instrument shall rotate on a tapered bearing surface or ball bearing races. All bearings shall move without binding or end-play throughout the temperature range from -25 °F to +120 °F.

Leveling screws. The leveling of the instrument shall be accomplished by movement of three leveling screws, which shall center the circular bubble in the vial. Leveling screws shall operate without binding, backlash, or end-play throughout the temperature range from -25 °F to +120 °F.

Tripod. The level shall be provided with a wide frame tripod having fixed or extension type legs. The tripod shall be constructed of hardwood that shall not bend, warp, or be affected by changes in temperature or humidity. The tripod shall be furnished complete, and include a tempered-steel shoe on the lower end of each leg, metal head, and head cover. The shoe shall be hardpointed, fastened to the leg, and equipped with a push-down spur. The head, or an adapter ring when furnished, shall be metal, and have threads to accommodate the level baseplate nonmagnetic material as the head, or of high impact plastic; when the level is mounted on the head, the cover shall be attached to the tripod for storage so that the cover is not capable of movement. When the head-protecting cap is removed and the legs are placed together, the overall length of the tripod shall be not less than 66 inches. A binding strap to fasten the legs together shall be attached to one leg.

Stadia constant. The instrument shall be mounted on a tripod or stand and leveled, and a graduated rod shall be located at a distance of 100 feet from the level. With the rod in the vertical position and carefully plumbed, the stadia interval shall be read by means of the stadia hairs. The stadia reading shall be multiplied by the specified stadia constant (see Reticle paragraph below). Variations of the stadia distance shall not exceed ±0.3 percent of the measured distance.

Reticle. The reticle shall consist of a pattern of opaque lines permanently registered on optical glass and mounted on a metal reticle cell. The reticle shall be arranged to eliminate distortion during collimation. Means for achieving collimation adjustments shall be

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positive and accessible. The reticle cell shall be adjustable for precise positioning of crosslines in the horizontal and vertical planes. Horizontal adjustment shall be by means of 3 or 4 capstan head screws that fasten the cell to the telescope and permit rotation of the cell. The reticle cell shall be provided with peg adjustments, a fitted sliding component, and adjustment screws for movement of the reticle in a vertical plane. Horizontal and vertical stadia lines shall be a minimum of 5 minutes of arc in length, parallel to the corresponding crosslines, and spaced at appropriate stadia intervals, from the vertical axis of the level to the rod. The rod shall be located at a predetermined distance greater than 100 feet or 30 meters from the level. Readings shall not result in errors that exceed 0.3 percent of the horizontal distance and shall not include allowance for the instrument constant. The width of all reticle lines shall be 3 seconds of arc +0.0 seconds -0.5 seconds.

Automatic compensator. Type II levels shall be equipped with an automatic compensator which will maintain a true horizontal level line-of-sight at all positions of the telescope and it will have a working range of not less than ±15 seconds of arc with a dampening capability of not more than 5 seconds of time to return to true level after the telescope is rotated.

Dampening mechanism. The compensator shall be provided with a dampening mechanism to stop oscillation and permit return and stability to the prism within 5 seconds.

Optical micrometer. The optical micrometer, when provided, will measure the interval between the horizontal reticule line and the nearest graduation of the rod directly to 0.0001 meter.

Horizontal circle. The horizontal circle, when provided, will be graduated at one degree intervals with the graduations numbered clockwise every 10 degrees. Angles shall be read through the microscope eyepiece on the optical scale directly to 10 minutes. Angles shall be estimated to the nearest minute and shall be accurate within ±1 minute.

Carrying case. The carrying case shall be constructed of metal, hardwood, glass-fiber-reinforced plastic laminate, or high impact plastic, contractor's option, with the inner liner form-fitted to the level. The case shall withstand severe operating usage and shall be blocked and padded to prevent damage when the instrument and case are dropped with a force equivalent to 10 times the weight subject to gravity. Each case shall contain desiccant adequate for the size of the container. The glass fiber and hardwood case shall have reinforced corners. The level shall be fastened within the case to prevent the instrument from falling out when the case is opened. The case shall be provided with locking catches. When constructed of plastic, the inner liner of glass fiber case shall be removable for cleaning. Fixtures shall be so installed inside the case as to prevent movement, and to protect repair parts and accessories against damage and breakage during storage and transit. A carrying strap or handle shall be fitted to the case to provide means for carrying the

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level in an upright position. The carrying case shall protect the level against damage from rainfall, sand, dust, shock, and vibrations during field use.

Accessories. The level shall be furnished with a lens-cleaning brush, screwdriver, adjusting pin, objective lens dustcap, eyepiece lens cover, sunshade, and waterproof cover.

Contractor certification. The Contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with Federal Acquisition Regulation, section 23.403, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packing, labeling, and marking shall be as specified in the contract or order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the Contractor.

Part or identifying number (PIN). The PIN to be used for levels acquired to this commercial item description is created as follows:

A52036 - Y - X - X
- X: Category 1 or 2
- X: Style: A or B
- Y: Type I or II
- A52036: Document identifier

Notes. The procuring agency should specify the preferred options permitted herein and include the following information in procurement documents:

1. Title, number, and date of this item description.
2. Type of level required.

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3. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.

FED-STD-376 is available from the Naval Publishing and Printing Service Office (NPPSO), Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

Preparing activity for this CID is: Commander, US Army Belvoir RD&E Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606.

MILITARY INTERESTS:

Custodians
Army - ME
Air Force - 99

Review activity
DLA-GS

User activities
Navy - MC, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-OSS
COM-NBS

Preparing activity:
Army - ME

Project 6675-0945

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